Task 5 - Header Files and Conditional Compilation

C Programming Assignment - Rural Electricity Billing System

Program structure

billing.h contains declarations and shared types; billing.c contains billing functions; main.c handles input and output; tariff.cfg contains configurable rates.

Include guard

#ifndef BILLING_H

#define BILLING_H

/* declarations go here */

#endif

Debug support

Use conditional compilation such as #ifdef DEBUG around debug logging. In a release build, debug statements can be omitted.

Compile and link

Compile billing.c and main.c separately with gcc -std=c11 -Wall -Wextra -Wpedantic -c, then link the object files using gcc main.o billing.o -o billing.

Header syntax

Angle brackets are normally used for standard headers such as stdio.h. Quoted includes are normally used for project headers. Compile-time debugging may reduce release executable size; runtime debugging is more flexible but can add overhead.

Important: illustrative tariff rates are used because the screenshot does not show the complete official tariff table. Replace them with your instructor's prescribed rates.